

Complex Physics: Midterm Assignment 1

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The system to be simulated is the 1D Ising model with N spins s_i on a ring. The spin-spin interactions are given by the (zero magnetic field) Hamiltonian

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j, \quad (0.1)$$

where $J > 0$ causes ferromagnetic coupling, which will be used in this assignment. The spin can take on the values

$$s_i = \begin{cases} +1 \\ -1 \end{cases} \quad (0.2)$$

In equation (0.1) the sum runs over every link between any spins, avoiding double counting. Every interaction is given the same strength J , which will be set to +1. Two geometries are simulated

1. Each spin is only connected to its nearest neighbours (nn), giving each spin two bonds (N bonds in total). This is the standard Ising model.
2. Each spin is connected to its nn, but also to another two random spins in the chain, meaning each spin is connected to four others. This effectively creates a new 1D system within the original system. The assignment of random next-nearest neighbours is only done once for a given system size

For the simulation we will use the Metropolis-Hastings algorithm at temperature T (or $\beta \equiv \frac{1}{k_B T}$). Starting from a random configuration with up- and down-spins of equal probability, the algorithm is as follows:

1. Choose a random spin to (potentially) flip.
2. Calculate the change in energy ΔE when going into the new configuration.
3. Assign the probability of transitioning to the new state as $w = \min(1, e^{-\beta \Delta E})$, i.e. 1 if $\Delta E \leq 0$ or $e^{-\beta \Delta E}$ if $\Delta E > 0$.
4. If $w = 1$: Transition. Else: Generate a random number between 0 and 1. If this is less than $e^{-\beta \Delta E}$, transition; if not, stay in the same configuration.

5. Repeat $n_{\max}$ times, and record the magnetisation $m \equiv \frac{1}{N} \sum_{i=0}^{N-1} s_i$ for every l 'th step, with l chosen such that 1000 samples will be taken.

After this, the mean of the magnetisation can be found from the recorded values. To exclude transient effects, a value n_0 is chosen, such that values of m recorded before this are not used in the average. If k values are included in the averaging, it is simply found as

$$\langle m \rangle = \frac{1}{k} \sum_{i=1}^k m_i \quad (0.3)$$

The mean magnetisation (or its absolute value) serves as the order parameter of the system. Therefore, it will be plotted for several values of T for each system type, and this will be used to investigate, whether the system undergoes a phase transition from order to disorder at some critical temperature T_c . In the following we measure temperature in units of $\frac{J}{k_B}$.

1

The mean magnetisation is plotted as a function of temperature. In this section, the system size will be set to $N = 1000$ for all simulations. To ensure that $n_{\max}$ and n_0 are large enough to exclude transient effects, the magnetisation can first be plotted as a function of the step in the algorithm. This can be seen for both systems at different temperatures in figure 1. I choose to set $n_0 = \frac{n_{\max}}{4}$ for all simulations, while varying $n_{\max}$. During the running of the algorithm, I will only save the value of the magnetisation at 1000 points, meaning around 750 points are used in the averaging.

From figure 2a we see that system A seems to order at very low temperatures, while the order parameter drops to zero for larger T . System B shows a different behaviour (figure 2b). It orders all the way up to a temperature $T > 2.5$, where the order parameter then drops dramatically to zero. This is a clear sign of a second order transition at a critical temperature T_c . Determining this temperature is the subject of the next section.

2

We now look at determining T_c for each system as the system size N goes to infinity (or $\frac{1}{N} \rightarrow 0$).

From the previous section, it seems that for system A, the critical temperature is less than 0.7, while for system B, it is in the range $1.8 \leq T \leq 3.1$. Thus, the temperature range is restricted to these values. System sizes of $N = 50, 250, 500, 750, 1000$ will be simulated.

The plots of $|\langle m \rangle|$ as a function of T can be seen for system A in figure 3 and for system B in figure 4.

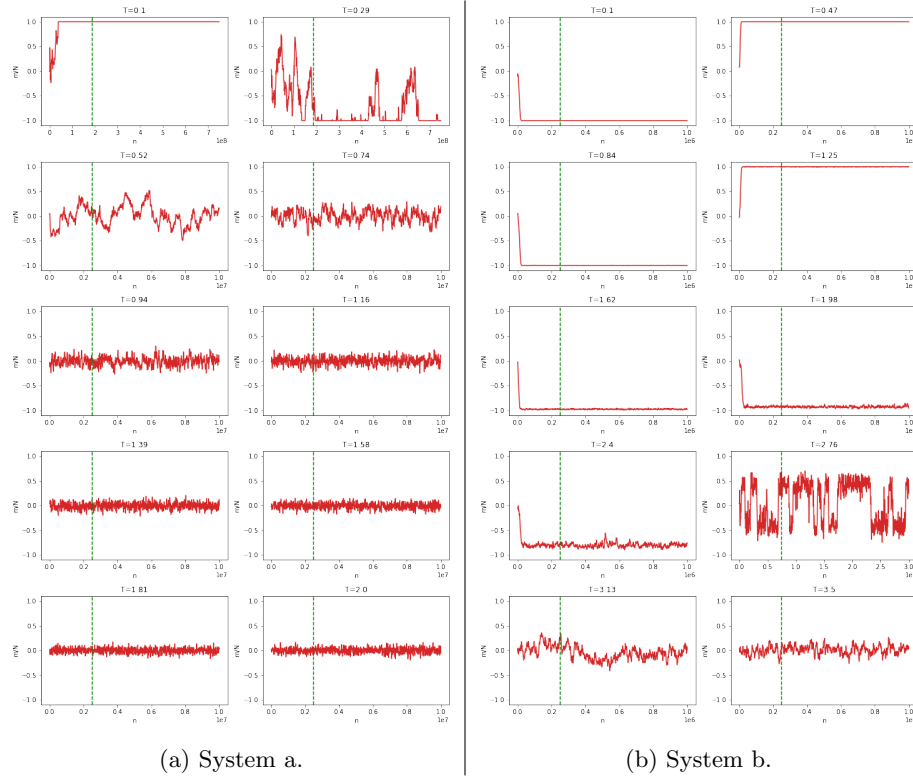


Figure 1: Examples of magnetisation as function of number of steps in the simulation. The dashed green line shows n_0 .

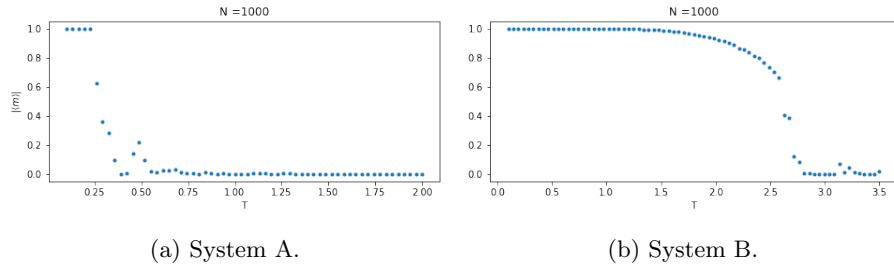


Figure 2: Absolute magnetisation as a function of temperature.

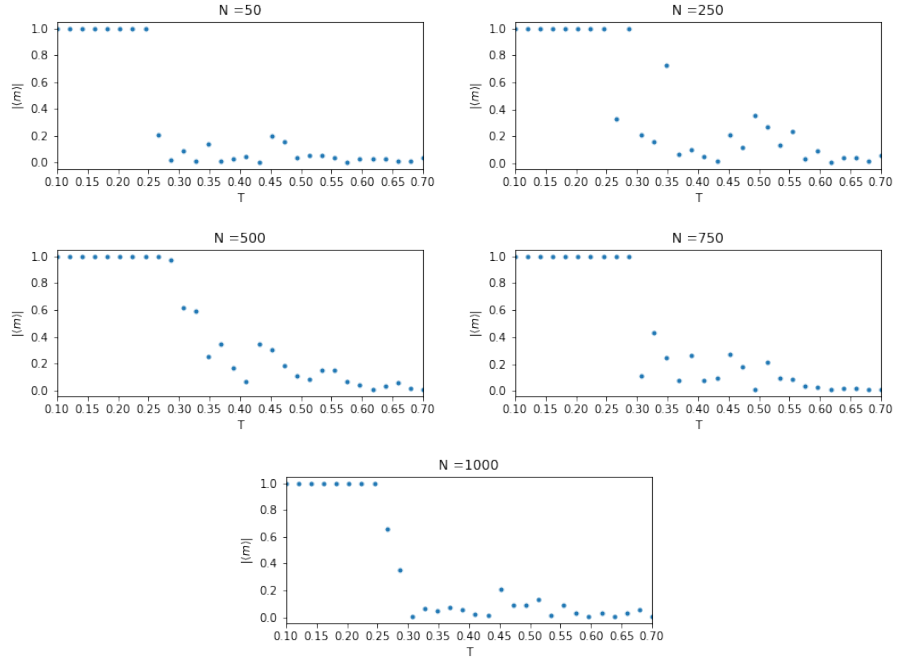


Figure 3: Absolute value of mean magnetisation as function of temperature for different sizes N of system A.

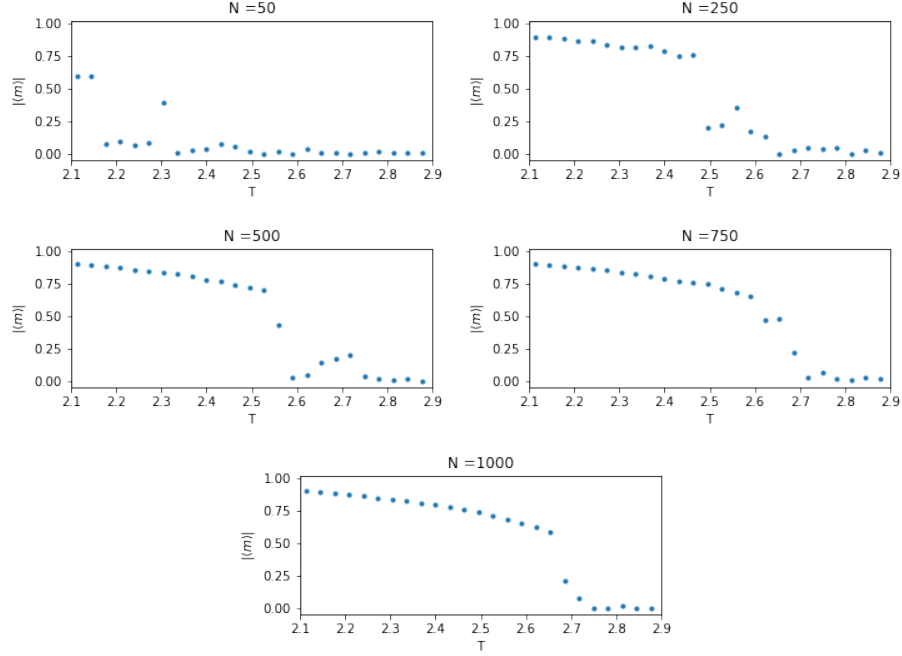


Figure 4: Absolute value of mean magnetisation as function of temperature for different sizes N of system B.

For system A the absolute magnetisation is seen to vary a lot, and determining a critical temperature is therefore difficult. Nonetheless, it seems that the magnetisation above $T = 0.3$ seems to drop off as the system size is increased. Based on this, an estimate of the critical temperature would be around $T_c^{(A)} = 0.3$.

For system B, a clear second order phase transition is seen. For larger system sizes, it can be seen that the magnetisation goes to zero around $T_c^{(b)} = 2.75$.

These are the critical temperature found by inspection from the magnetisation plots. However, there are a lot of possible sources of error, and especially $T_c^{(A)}$ seems off, as we expect the 1D Ising model to exhibit disorder at all finite temperatures. The current program used for the simulation has not been fully optimized, and simulating larger systems is therefore costly. It would be interesting to continue working on this, such that larger systems can be investigated in order to ascertain, whether the ordering of system A persists, even though it should not.

One might expect the magnetisation at a given temperature to increase with decreasing system size, as the coherence length becomes a larger fraction of the

finite system size. However, this is not always seen, for example for $N = 50$ for system B. The fact that magnetisation sometimes seems to decrease for decreasing system size could be due to the order parameter flipping sign during the simulation, as can be seen for the $T = 0.29$ simulation of figure 1a. This could happen more easily if the system is small, since there are less spins to flip. Flipping the order parameter will lead to a lower mean value, even if the system has ordered to an extent. This also explains the large fluctuations in magnetisation seen especially for small system sizes.

It is also interesting to note that $n_{\max}$ generally had to be set higher for system A to exclude transient effects. This shows that although system A orders for small temperature, this order does not dominate as quickly as for system B.

3

When finding the state of a Ising model system, the free energy $F = \langle E \rangle - TS$. For system A, energy increases when neighbouring spins are oriented antiparallel, leading to an energy gain of $E_{\text{wall}} = 2J$ relative to the case of parallel spins. This can create a domain wall. This domain walls increase the entropy (placing x domain walls at N sites), and so these compete for the minimisation of the free energy. Assuming $x \ll N$ we get

$$F = 2Jx - k_B T \ln \left(\frac{N!}{x!(N-x)!} \right) \approx 2Jx - k_B T x \ln \left(\frac{N}{x} \right) \quad (0.4)$$

Defferentiating with respect to x and finding the minimum $\frac{\partial F}{\partial x} = 0$, we get a rough estimate of the mean size of domains N/x as;

$$\langle \text{domain size} \rangle \simeq \exp \left(\frac{2J}{k_B T} \right). \quad (0.5)$$

Even though this result is an approximation, similar results can be derived by other means. This shows that the coherence length drops off exponentially, and for a system with $N \rightarrow \infty$ there will be no long ranged order. The fact that some order was seen for low temperatures in the simulations of this system are likely due to the still-finite system size. Computationally, the low number of neighbours means $\Delta E_{\max} = 4J$ small, meaning $w = e^{-\frac{\Delta E_{\max}}{k_B T}}$ is large (the transition probability is large for any ΔE). However, if T is also sufficiently small, the size of the energy change becomes important again, and states of low ΔE are more likely to be chosen. We can indeed compare with the 1D Ising model correlation length (equation 2.53 in the lecture notes)

$$\xi(T) = -\frac{1}{\ln(\tanh(\frac{J}{k_B T}))} = -\frac{1}{\ln(\tanh(T^{-1}))}. \quad (0.6)$$

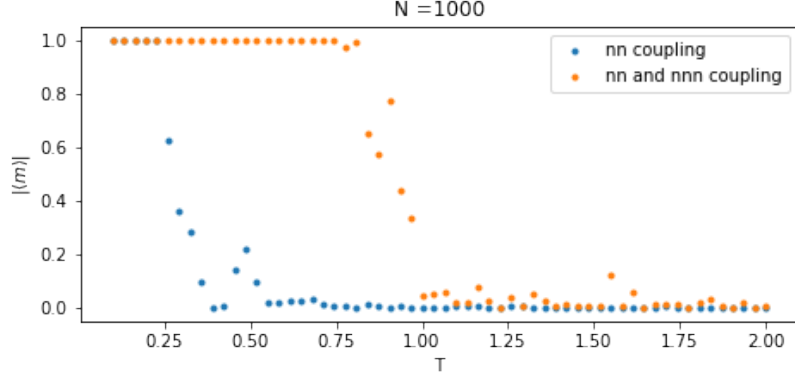


Figure 5: Comparison of simulations of $N = 1000$ systems: One system includes only nearest neighbour interactions, while the other also includes next-nearest neighbour interactions.

For $T = 0.1$, $\xi \approx 24 \cdot 10^7$, considerably larger than the system sizes used here.

This also gives a qualitative explanation of why system B orders at much higher temperatures. We now have $\Delta E_{\max} = 8J$ and a higher correlation between spins at different sites in the chain. Something similar might be true for a 1D Ising system with nearest and next-nearest neighbour coupling, where domain walls would cause a larger energy increase. This is shown in figure 5, where it can be seen, that the system which includes next-nearest neighbour interactions orders at higher temperatures.